KINGSHUK DHOLAKIA

kingshuk.dholakia@gmail.com • LinkedIn

(0415) 571-286 • Camperdown, NSW

Game Developer

Detail-oriented Game Development graduate with proven experience in Unity and C# through full-cycle game projects, from concept to showcase recognition. Knowledge of SOLID principles and commonly applied design patterns such as Singleton, Factory, and Adapter. Familiar with Generics, Interfaces, Sealed and Partial Classes, Events, Delegates, Attributes, and Reflection for scalable code architecture. Capable of applying mathematical modelling, probability, and logic validation to support accurate gameplay systems and simulations. Skilled in debugging, troubleshooting, and performance profiling to identify and resolve issues across builds. Ability to prepare structured documentation and contribute to peer code reviews to support testing and quality compliance. Experienced in integrating art assets, audio, and UI/UX elements into interactive systems, troubleshooting, and supporting game logic. Staying up to date with industry advancements in game engines, digital content creation software, and tools.

Areas of Expertise

- Gameplay Programming
- Game Mechanics Design
- Collision Detection
- Level Design & Audio Design
- Team Leadership & Collaboration
- Object-Oriented Programming (OOP)
- Project Management
- Animation Integration
- Storyboarding & Game Balancing
- Visual Effects Integration
- Code Architecture & Debugging
- Physics Simulation
- UI/UX Implementation
- Math Modelling for Games
- Data Structures & Algorithms

Technical Skills: Unity | Trello | Piskel | Github | Canva | Miro | C# | JavaScript

Education

Bachelor of Science in Games Development, University of Technology Sydney | 2021 - 2025

Major Courses: Game Design Studio 1 | Game Design Studio 2 | Fundamentals of Interaction Design

Key Projects

Solo Developer & Product Lead | Project: Everyday Strong In Progress

Tools & Technologies: React Native • Expo • TypeScript • Claude Code • GitHub

Designing and building an offline-first mobile movement app for adults aged 50–70, organised around real-life goals such as getting up off the floor, carrying shopping, and climbing stairs. Progress is narrated through self-comparison rather than streaks or scores, with an optional circle for sharing milestones with family, and accessibility built in as the baseline.

Key Contributions:

- Defined a strict wellness safety boundary (motivate and educate, never assess or diagnose) and enforced it in CI with an automated copy gate that blocks clinical or judgemental language across the app.
- Directed development with an AI coding agent through phased pull requests, specifying each phase upfront and reviewing every diff before merge, reaching a feature-complete build with 113 passing tests.
- Led evidence-driven pivots on the movement demonstrator, from 3D animation to a 21-movement illustrated library to a schematic 2D figure, selecting the final renderer through a measured performance comparison of three approaches.

Lead Game Developer & Project Manager | Project: Spectrum Ops - Rescue Mission August 2024 – November 2024

Tools & Technologies: Unity • Miro • GitHub • Audio Editing Tools (custom + sourced)

Directed a 3-member development team as both Lead Developer and Project Manager by organising tasks in Miro, setting milestones, and maintaining accountability through structured check-ins to ensure consistent progress under strict deadlines.

Key Contributions:

- Built the majority of Unity gameplay systems, including core mechanics and interactive features, by translating design ideas into functional code that balanced playability with performance.
- Oversaw the creative and technical audio pipeline, composing original tracks, designing sound effects, and refining sourced assets to establish an immersive and professional soundscape aligned with the game's theme.
- Guided the project from concept to delivery in a highly constrained timeframe by earning top academic marks and formal commendation from the Tutor and Subject Coordinator for leadership, technical execution, and overall game quality.

Game Developer & Artist | Project: Dungeon Golf March 2024 – May 2024

Tools & Technologies: Unity • Trello • GitHub

Created all enemy sprites and visual assets from scratch in Unity, iterating through multiple design revisions and incorporating peer feedback to achieve a polished and cohesive art style. Implemented smaller mechanics for seamless integration between art, design, and functionality.

Key Contributions:

- Collaborated with a multidisciplinary team for the first group game project, using Trello to manage tasks, coordinate progress, and build team accountability throughout the development cycle.
- Achieved the highest possible grade, which secured an invitation to the UTS Games Showcase, where the game received direct praise from Riot and Ubisoft developers for creativity and execution.

- Earned wider recognition, including a Twitter shoutout from the Game Curator of the Powerhouse Museum, Sydney, highlighting the project's innovation and presentation quality.

Solo Developer | Project: Procedural Dungeon Generator March 2024 – May 2024

Tools & Technologies: JavaScript • HTML5 Canvas

Built a standalone level-generation tool alongside Dungeon Golf to give its roguelike runs varied, playable layouts. Implemented three generation algorithms in plain JavaScript with no engine or libraries, animating each one step by step so its output could be inspected, compared, and tuned.

Key Contributions:

- Implemented Binary Space Partitioning for structured rooms and corridors, cellular automata for organic caves, and drunkard's walk for winding tunnels, with live parameter tuning and instant regeneration for side-by-side comparison.
- Used the step-by-step visualisation to expose chokepoints and spaces too open for golf-shot traversal, directly informing how Dungeon Golf's levels were structured.
- Turned procedural generation from a black box into a visible, debuggable process, producing a reusable engine-free tool that outlived the project it was built for.

Languages: English, Hindi & Gujarati: Fluent